MEDIANTS AND DOUBLE ANGLES ON ONE PAGE

Adding fractions the Farey way

Adding numerators and denominators adds vectors, while the angle-addition formula multiplies them.

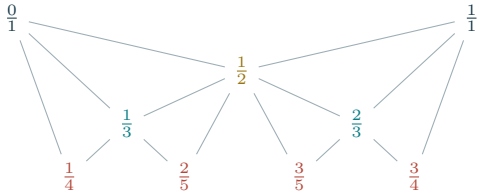
THE RULE

The mediant lands between, and only between

The mediant of two fractions adds tops and bottoms separately,

$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d} = \frac{b\cdot\frac{a}{b}+d\cdot\frac{c}{d}}{b+d}$$

so it is the average of the two values weighted by their denominators, and it always lies strictly between them. Two fractions are Farey neighbours when $|ad-bc|=1$; then their mediant is the simplest fraction between them, and starting from $\frac{0}{1}$ and $\frac{1}{1}$ and taking mediants of neighbours produces every fraction in $[0,1]$ exactly once, in lowest terms. The mediant of a fraction with itself, $\frac{2a}{2b}$, is the same number, so mediants can split an interval but can never double anything.



Each new fraction is the mediant of the two neighbours it is joined to.

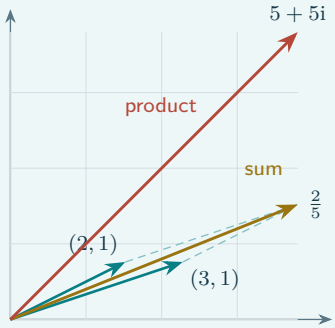
TWO WAYS TO COMBINE SLOPES

Mediants add vectors, angle addition multiplies them

Read the slope $\frac{p}{q}$ as the vector (q,p), or the Gaussian integer $q+\mathrm{i}p$, at angle $\arctan\frac{p}{q}$. Adding two vectors gives the mediant, whose angle lies between theirs, pulled towards the longer vector; multiplying them as complex numbers adds the angles exactly:

$$\frac{p}{q}\oplus\frac{c}{d}=\frac{p+c}{q+d}, \qquad \tan(\alpha+\beta)=\frac{pd+qc}{qd-pc}.$$

With $\frac{1}{2}$ and $\frac{1}{3}$, the sum $(2,1)+(3,1)=(5,2)$ gives the mediant $\frac{2}{5}$ at 0.3805 rad, between 0.3218 and 0.4636, while $(2+\mathrm{i})(3+\mathrm{i})=5+5\mathrm{i}$ gives slope 1, the classic identity $\arctan\frac{1}{2}+\arctan\frac{1}{3}=\frac{\pi}{4}$.



The sum lies between the two; the product swings out to $45°$.

DOUBLING

Squaring doubles the angle; a mediant with itself stays put

Squaring $q+\mathrm{i}p$ gives $(q^2-p^2)+2pq\,\mathrm{i}$, the traditional double-angle formula in fraction form, and its three numbers are a Pythagorean triple. The two operations divide the work: mediants of Farey neighbours reach every slope $\frac{p}{q}$, and doubling the angle of each slope turns it into a triple.

	mediant	angle addition
on vectors (q,p)	add: $(q+d,\ p+c)$	multiply as $(q+\mathrm{i}p)(d+\mathrm{i}c)$
on angles	lands strictly between	adds exactly
combined with itself	$\frac{p}{q}\oplus\frac{p}{q}=\frac{p}{q}$	$\tan 2\alpha=\frac{2pq}{q^2-p^2}=\frac{2t}{1-t^2}$
depends on the form of the fraction	yes, $\frac{1}{2}\oplus\frac{1}{3}\neq\frac{2}{4}\oplus\frac{1}{3}$	no, only on the value
what it builds	every fraction, once, by Farey	every Pythagorean triple, by doubling

slope $\frac{p}{q}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{2}{5}$	$\frac{3}{5}$
doubled: $q^2-p^2,\ 2pq,\ p^2+q^2$	$3,4,5$	$8,6,10$	$5,12,13$	$15,8,17$	$7,24,25$	$21,20,29$	$16,30,34$

FRACTIONAL RADIANS

Rational angles or rational slopes, not both

Taking mediants of angles measured in radians gives a weighted mean angle: $\frac{1}{2}$ rad $\oplus$ $\frac{1}{3}$ rad $=$ $\frac{2}{5}$ rad, the average of 0.5 and 0.333 weighted $2:3$. But Lambert proved in 1761 that $\tan r$ is irrational for every rational $r\neq 0$, so a rational angle never has a rational slope; Farey arithmetic can live on the angles or on the slopes, and only the slopes lead to triples.

WHERE MEDIANTS EARN THEIR KEEP

Best approximations and gear trains

The convergents of a continued fraction are reached by repeated mediants, which is why $\frac{22}{7}$ and $\frac{355}{113}$ appear on the Stern–Brocot path to π; a clockmaker choosing gear teeth to approximate a ratio walks the same tree. A season's success rate is the mediant of its parts, weighted by attempts, so one player can lead in every half-season yet trail overall, which is Simpson's paradox.

Conventions. $\oplus$ is the mediant and slopes $\frac{p}{q}$ are in lowest terms; when p,q are both odd the doubled triple halves to a primitive one, as $8,6,10$ becomes $4,3,5$.